# Cluster Size vs $E_T$ — Short Report

PPG12

April 16, 2026

**What is plotted:** per-cluster mean tower count $n_{\text{owned}}$ (from `cluster_ownership_array`) and bounding-box widths $\Delta i_\eta$, $\Delta i_\phi$ vs reco $E_T$, for data, photon MC (inclusive), jet MC (inclusive), single vs double interaction, and the split (`CLUSTERINFO_CEMC`) vs legacy no-split (`CLUSTERINFO_CEMC_NO_SPLIT`) cluster containers. Selection: $|\eta_{\text{reco}}| < 0.7$ only (no BDT / NPB / iso). DI mixing uses bin-by-bin $\langle y \rangle_{\text{mix}} = (1-f)\langle y \rangle_{\text{single}} + f\langle y \rangle_{\text{double}}$ with $f = 0.079$ (1.5 mrad) or 0.224 (0 mrad).

**Key findings**

- **Data is ∼1–3 towers larger than jet MC in the split container** (max gap +2.7 at $E_T = 25$ GeV), but only ∼1 tower larger in no-split. The excess is $\eta$-asymmetric (visible in $\Delta i_\eta$, absent in $\Delta i_\phi$) and diffuse, not clumped — consistent with noise/threshold contamination rather than real secondary showers.
- **Split clusterizer removes ∼30% of jet cluster towers** at $E_T > 15$ GeV and ∼1–4% of photon cluster towers (Fig. 2).
- **DI broadens jets ∼4× more than photons**: pure-DI shift is +0.5 to +1.3 towers (jet) vs +0.2 (photon). 1.5 mrad mix is indistinguishable from single; 0 mrad mix is visible.

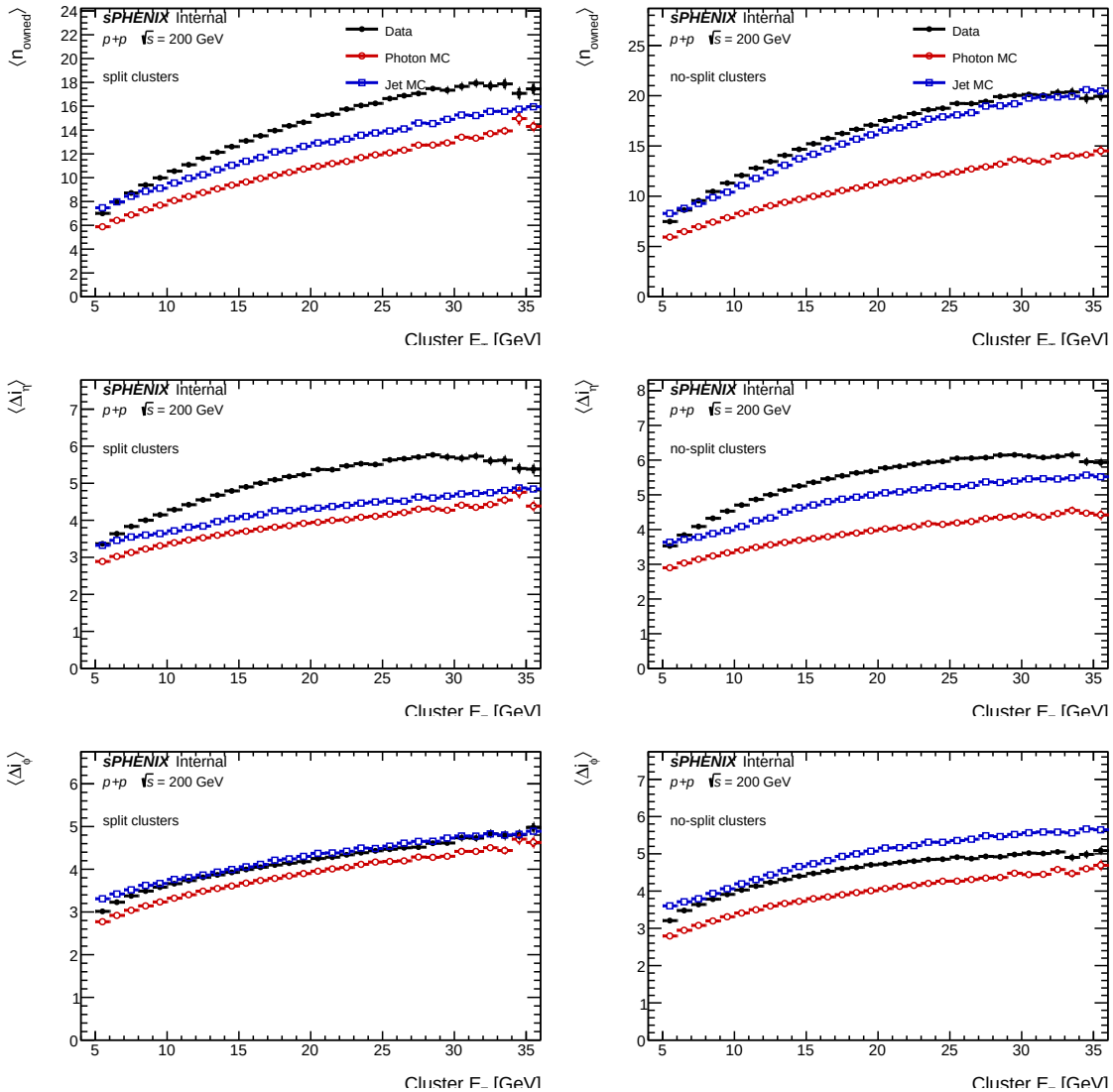


Figure 1: Data (black), pooled photon MC (red), pooled jet MC (blue) for $n_{\text{owned}}$, $\Delta i_\eta$, $\Delta i_\phi$ vs $E_T$ in the split (left) and no-split (right) containers.

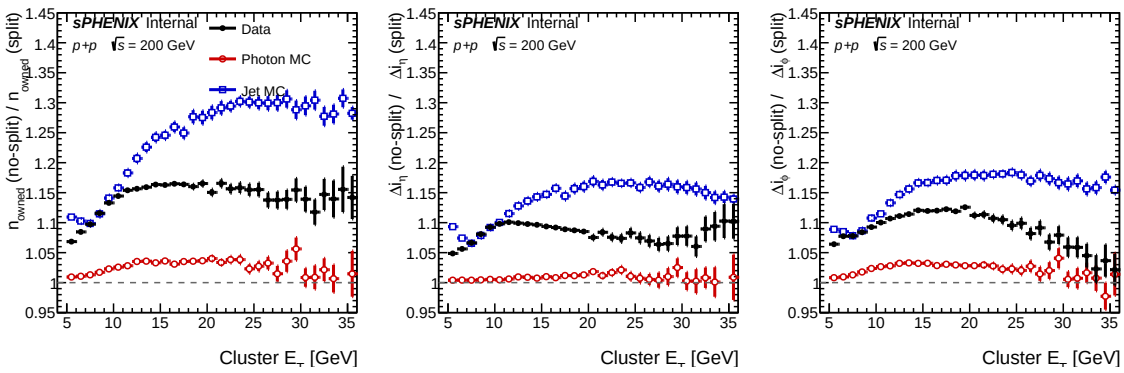


Figure 2: Ratio of mean cluster size in no-split / split for each category and metric. Photon MC ≈1; jet MC rises to ≈1.30 at high $E_T$; data sits below jet MC.

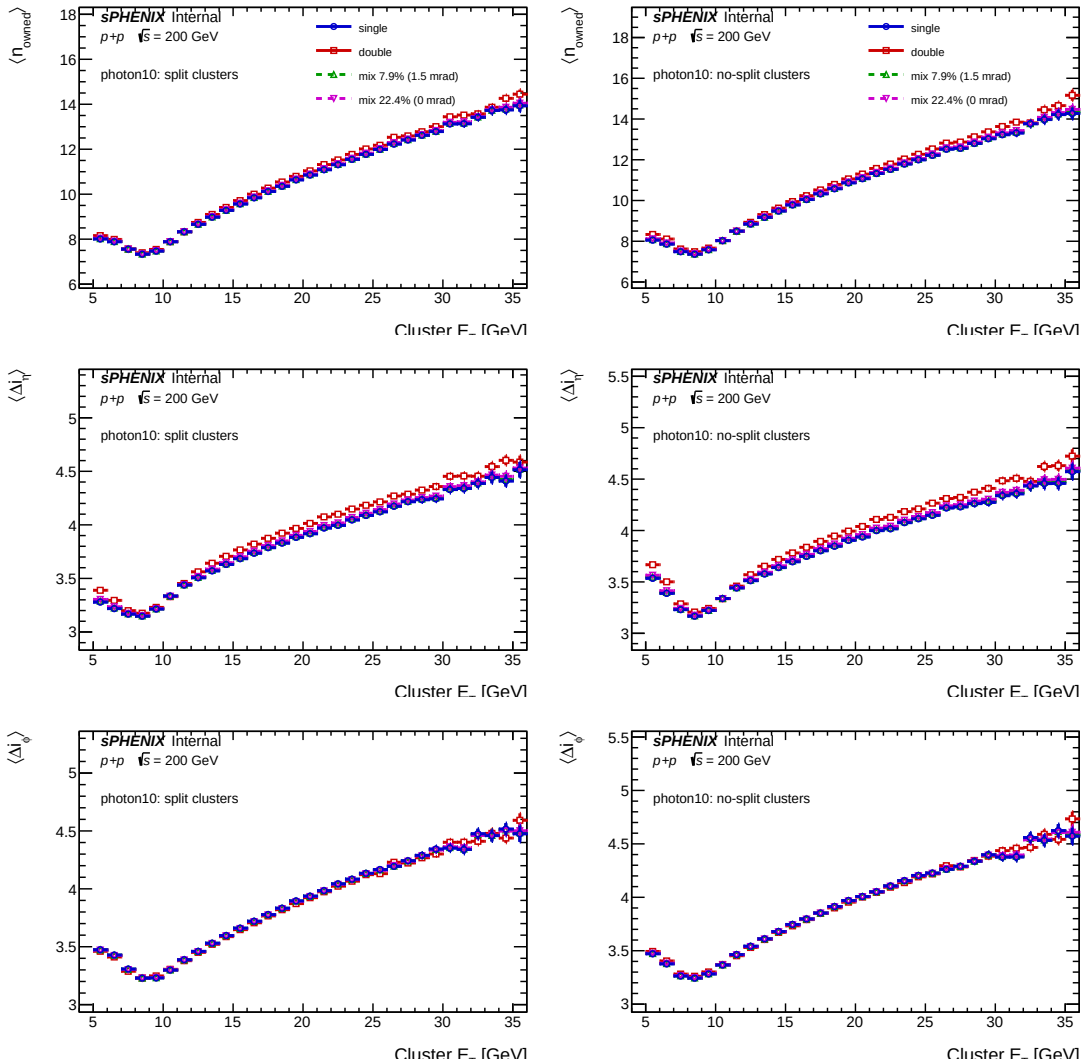


Figure 3: Photon DI: single (blue) vs double (red) vs mixed 7.9% (green) vs mixed 22.4% (magenta), for `photon10_nom` and `photon10_double`, split (left) and no-split (right).

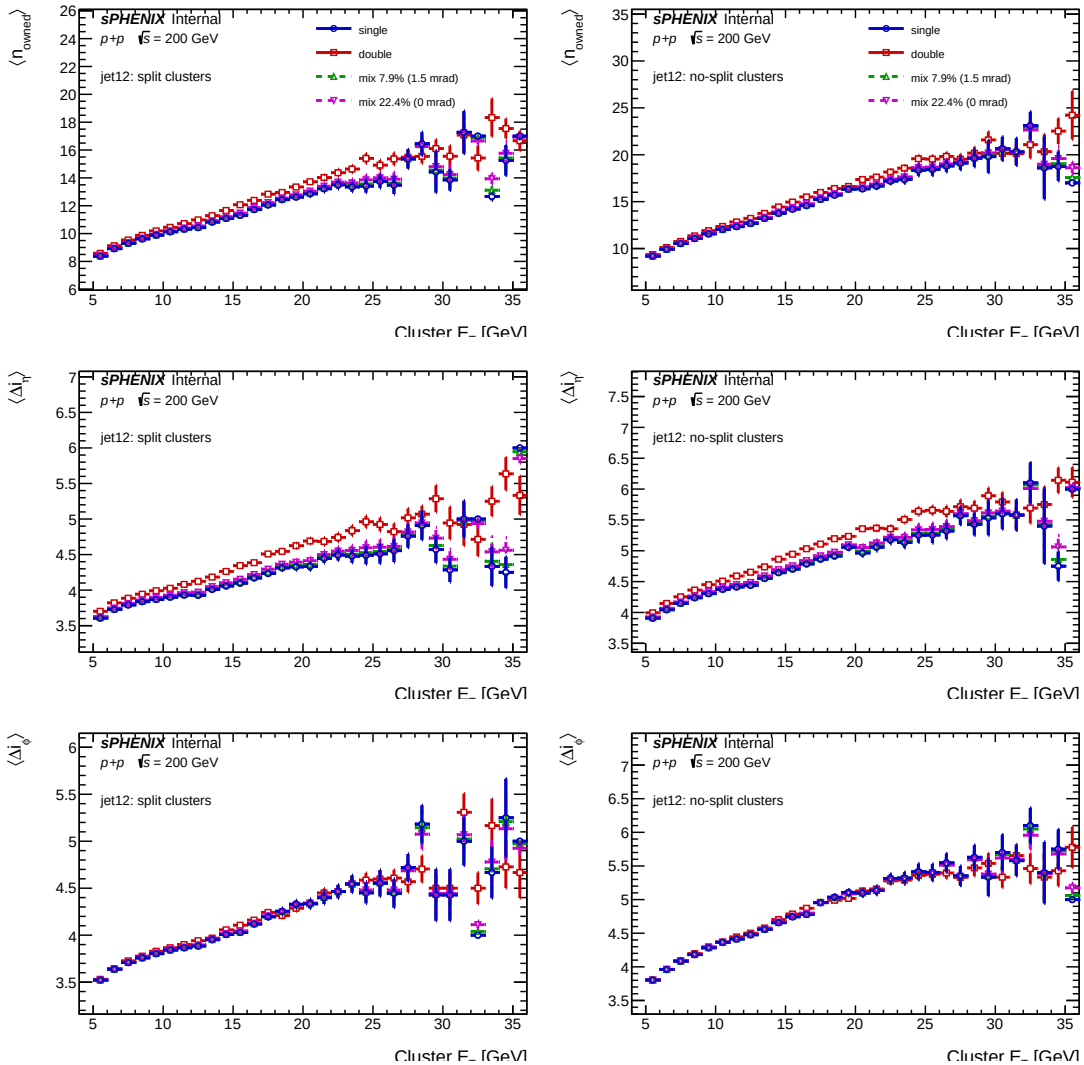


Figure 4: Jet DI: same format for `jet12_nom` and `jet12_double`. DI effect on jets is several times larger than on photons.

**Full report:** `cluster_size_vs_et.pdf` has the sample inventory, detailed per-figure observations, and recommended follow-ups (noise/threshold study, BDT training implications).